Detecting Machine-Generated Academic Text Using Supervised Learning

Dissertation

Course: MSc Data Science

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Abstract

The growing popularity of the text written by AI, especially in the academic sphere, brings new challenges to academic integrity. This dissertation examines the possibility of identifying machine-generated academic writing with supervised learning, such as both stylometric analysis and transformer-based deep learning models. Four classifiers including Logistic Regression, Support Vector Machine, BERT, and RoBERTa are trained on a dual-domain essay dataset containing human-and machine-written essays to evaluate the performance of the classifiers. To have a robust comparison of performance of each model, they are evaluated in terms of accuracy, macro F1 score, ROC-AUC, and confusion matrices.

Besides model-based classification, a pre-trained GPT-2 model is used to provide an unsupervised benchmark in the form of a perplexity-based baseline. The study also performs a cross-domain generalisation test that measures the performance of detection models trained on academic data applied to AI-generated texts in different domains. Regarding the considered models, RoBERTa outperforms the rest in terms of its overall performance, and the results on all evaluation metrics are better. The results point to the practicality of using explainable stylometric features, together with high-performance transformer models, to reliably identify AI-generated scholarly writing.

The study will be of value to the emerging area of AI-authorship detection, and will help educators and educational institutions struggling with the issues of generative AI in academia by showing how effective, interpretable, and scalable solutions to the problem can be applied.

Acknowledgements

I would like to thank my supervisor **Dr. Bence Borda** a lot because of his constant patience, humility, and support with this research. His valuable comments, his timely availability, and his unbreakable encouragement were pivotal towards completion of this project. There was no one a more committed mentor could have been.

I would like to extend my profound gratitude to the University of Sussex that enabled me to work on this paper by availing the resources, tools, and research environment to carry out this work, which includes access to research labs, and high-performance computing units.

I also extend my appreciation to all the teaching staff of the MSc Data Science program, especially **Prof. Julie Weeds** and **Dr. Jeff Mitchell**, whose exceptional instruction in Advanced NLP and Applied NLP modules equipped me with the theoretical foundation and practical skills essential for this dissertation. Their passion for natural language processing inspired me to explore this field further.

Finally, to my family and friends, thank you for your endless patience and encouragement during this challenging yet rewarding journey.

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1. Introduction

1.1 The Rise of AI-Generated Text in Education

Artificial intelligence has been developing amazingly in language generation in recent years. Such tools as GPT-2 and GPT-3 can be used to create fluent and coherent text in a variety of subjects, and in the correct context. These models have been trained using large amounts of data, and can create text which can be hard to differentiate between one written by a human being. Although these tools hold promising prospects in areas such as journalism, education, and customer care, they present scientific questions of concern especially in the academic circles.

Among the most urgent ones is the application of AI to generate student essays. Since the material is not copied anywhere. This poses a very dangerous threat to academic integrity because institutions depend on written tests to gauge the student knowledge and effort.

1.2 Motivation for the Research

The technical problem of identifying AI-written essays is not only technical, but also a larger question of fairness in education. Making use of the AI tools to write assignments would give those students an unfair advantage, and yet others who do not break the rules are held to the same standard. Currently, colleges and teachers do not have enough means of identifying such instances.

Academic writing is structured and written in a different way to informal or conversational text. This leaves the door open to the possibility that even fluent machine-generated essays may contain some pattern that is characteristic of them and differentiates them somewhat with student writing. With machine learning being able to detect these patterns, systems can potentially be created that will allow institutions to identify the suspicious submissions and react to this new type of academic misconduct.

1.3 Aims and Research Questions

The primary objective of the given research is to investigate the possibility of supervised learning to identify the artificial intelligence generated academic essays. The project contrasts two general categories of models: the classical machine learning classifiers (stylometry), and new deep learning models (transformer-based). The simpler, zero-shot using language model perplexity is also provided as a baseline.

Instead of suggesting completely novel algorithms, the study explores the question of whether popular and well-established tools like logistic regression, support vector machines, and BERT/RoBERTa models fine-tuned to the task are enough to solve the problem effectively. This makes it possible to have more reproducible results and simpler to use in practice during the study. A second set of data concerning another domain is used to test the generalisability of the approach, based on the most effective model.

The research is guided by the following questions:

- Can features based on writing style help tell the difference between human-written and AI-generated academic essays?
- How well do fine-tuned transformer models such as BERT and RoBERTa perform in detecting AI-generated text?

- Is perplexity, as measured by a pre-trained GPT-2 model, a useful baseline for identifying machine-generated content?
- Can a model which trained on academic essays generalise to other types of AI-generated text beyond the essay domain?

By answering these questions, the study aims to show whether existing detection tools and standard supervised learning techniques are sufficient for identifying AI-written academic work. The findings are intended to support educators and researchers in responding to the growing use of generative AI in higher education.

1.4 Structure of the Dissertation

This dissertation is organised into eight chapters. Following this introduction, Chapter 2 reviews the relevant background and past research on text classification, stylometry, and detection techniques for machine-generated content. Chapter 3 presents the data sets collected and applied to this project and describes the data preprocessing procedures and data exploration. Chapter 4 describes the methodology, including the development, training, and testing of the stylometric and transformer-based models, unsupervised perplexity baseline, and the cross-domain testing protocol. Chapter 5 provides the results, the model performances and additional analyses. Chapter 6 will explain the most important findings, explain the model behaviour, compare the results with the other literature and point out the limitations of the study. Lastly, Chapter 7 is the conclusion of report that summarises the key contributions, and limits. There is a full list of references at the end.

2. Background and Related Work

This chapter provides a short summary of the main ideas and techniques behind the project. It also discusses stylometry, language modeling and existing approaches to identify machine-generated text. The aim is not to give a full review of the field, but only the essential information needed to understand the choices made in this dissertation.

2.1 Stylometric Analysis

Stylometry is the study of writing style. It has been applied in problems like authorship identification and plagiarism detection. Stylometric features are patterns in text that reflect how someone writes. These include things like variance in sentence length, vocabulary richness, punctuation use, and how often different types of words (like nouns or verbs) appear.

In this project, stylometric features were used to train simple machine learning models. The tools that were used to extract these features:

- spaCy: a Python library for analysing the structure of sentences
- NLTK (Natural Language Toolkit): a toolkit for text processing
- TextStat: a utility to gauge the ease or difficulty of reading a piece of text

The features were then transferred to classifiers such as logistic regression and support vector machines. These are standard models that are fast to train and easy to interpret.

2.2 Transformer-Based Models

Transformer models will be a recent and very strong development in natural language processing. They operate by focusing on every section of a sentence simultaneously, and not reading word after word. This makes them a lot more competent in grasping the meaning of a sentence as a whole.

The models of the main transformer utilized in this project were BERT (Bidirectional Encoder Representations from Transformers) by Devlin et al. (2019) and RoBERTa (A Robustly Optimised BERT Pretraining Approach) by Liu et al. (2019). The two models have identical fundamental structure on the basis of the transformer encoder and only vary in their pretraining approaches. BERT is pre-trained with masked language modelling and next sentence prediction on large corpora of English. Instead, RoBERTa does not use the next sentence prediction task, employs dynamic masking, and is trained on much larger amounts of data and with greater compute.

In the project, both models were optimized to perform the binary classification task to identify human- and AI-written academic essays. Each of the models employed used their respective tokeniser to tokenise the text inputs and each essay was truncated or padded to be of a fixed sequence length to fit the respective model requirements. Fine-tuning entailed adjusting model weights using the training data whilst evaluating the training on a validation set to avoid over-fitting.

Transformer models are powerful but can also be hard to explain and computationally expensive. Their results are compared to simpler models to see whether the extra complexity is justified.

2.3 Perplexity-Based Detection

Perplexity is a way to measure how well a language model can predict the next word in a sentence. If the model is very confident about what word comes next, the perplexity is low. If it is uncertain, the perplexity is high. The idea behind using perplexity for detection is simple: a language model like GPT-2 will find its own text more predictable than human-written text. So, if a passage has low perplexity under GPT-2, it might be machine-generated.

This method does not require training a classifier, which makes it useful as a baseline. However, it also has limits. It is sensitive to text length and formatting, and it assumes access to the model that might have generated the text. These issues are discussed in later chapters.

2.4 Related Work and Gaps

Machine-generated text detection now is an active research field, especially as it has become more common to encounter powerful large language models (LLMs) like GPT-2, GPT-3, and ChatGPT. Initial research in the field frequently used stylometric analysis, i.e. the application of hand-crafted features including sentence length, word usage, punctuation frequency and syntactic patterns to identify authorship. In authorship attribution, Stamatatos (2009) provides an extensive survey of stylometry, as well as how it has been applied in detecting plagiarism and in forensic linguistics.

With the advancement of neural language models, researchers moved on to classifiers based on deep learning that learns implicit representations of writing style. As an illustration, Ippolito et al. (2020) trained the BERT-based models to classify between the news articles that are written by humans and machines. Likewise, Gehrmann et al. (2019) have introduced GLTR (Giant Language

model Test Room), which takes visualizations of word probabilities in a language model and assists humans in detecting text that was generated. Although helpful, GLTR is too reliant on human interpretation and cannot be scaled to be automatically detected in real-time systems.

Mitchell et al. (2023) proposed DetectGPT, a zero-shot detection approach that is realized by text perturbation. This method approximates the log-likelihood surface of the data around a sample text to determine whether or not the text is within a region most likely to be occupied by model-generated outputs. Despite these good results, DetectGPT requires access to the probabilities of the underlying model tokens, constraining its use cases in examinations or third-party detection applications.

Despite these innovations, there are three critical gaps in the current literature:

1. **Domain Bias on Informal Texts:** A lot of current detection research uses datasets of informal or general-purpose text e.g., Reddit comments, online reviews, or news headlines. These have a lot of differences with academic writing that is more lengthy, structured and is linguistically formal. Ippolito et al. (2020) admit that the issue of generalization across domains continues to be a significant problem with a detector trained on one type of text not working on another one.

2. **Limited Attention to Educational Contexts:** There is insufficient coverage of the area of academic integrity in learning particularly during online examination and coursework in the detection literature. Although Turnitin and GPTZero provide commercially-available tools, there is a limited amount of peer-reviewed research which has specifically tested detection systems against publicly-available student-written essays.

3. **Excessive use of Model-Specific Assumptions:** Some of the techniques, including DetectGPT or watermarking techniques implemented internally by OpenAI, rely on knowing the model used to generate the work or the internal representations of the model. This cannot be practically applied to most real-life educational contexts where generated text can be authored by unknown models or models that belong to proprietary entities.

This project addresses those gaps by:

- A specific focus on academic writing as opposed to informal or general-purpose text.
- Evaluate the performance of the models via two publicly available datasets across domains.
- Contrast of two radically different methods: the classical stylometric classifiers and transformer-based language models.
- A new perplexity-based diagnostic analysis to explain model behaviour and aid in the decision of the classifier has been included.
- By focusing on transparency and interpretability, and explaining in a way that is accessible to non-technical experts like educators and policymakers

3. Data

This chapter introduces the datasets that will be used in the project, the preprocessing and cleaning procedure, and also gives an outline of the exploratory data analysis (EDA) undertaken. It also defines how the data was partitioned into training, validation and test. The approaches were all

made to facilitate the development of reliable models without data leakage and with the preservation of natural structure of academic text.

3.1 Source Datasets

3.1.1 Kaggle LLM Detect AI Text Dataset

The primary data source of the present project is the **LLM Detect AI Text Dataset** posted on **Kaggle** by Thite (2023). It has **29,145 essays** written in English, of which each is marked either as being written by a human or by an AI. The student essays are written by humans and the AI essays are produced by large language models- GPT-3, in this case. The data is available in CSV format in two columns:

- text: the full content of the essay
- generated: a binary label (0 = human, 1 = machine)

The dataset has **17,508** AI-generated essays and **11,637** human-written essays. The reason I chose this data set is that it deals with academic writing in particular. The style and form of the essays are typical of the work of students, so it can be used in training models that could recognize AI-generated assignments in the educational context. There is a slight imbalance in the class distribution with a few more human essays as compared to AI. The figure 1a presents the proportion of each class.

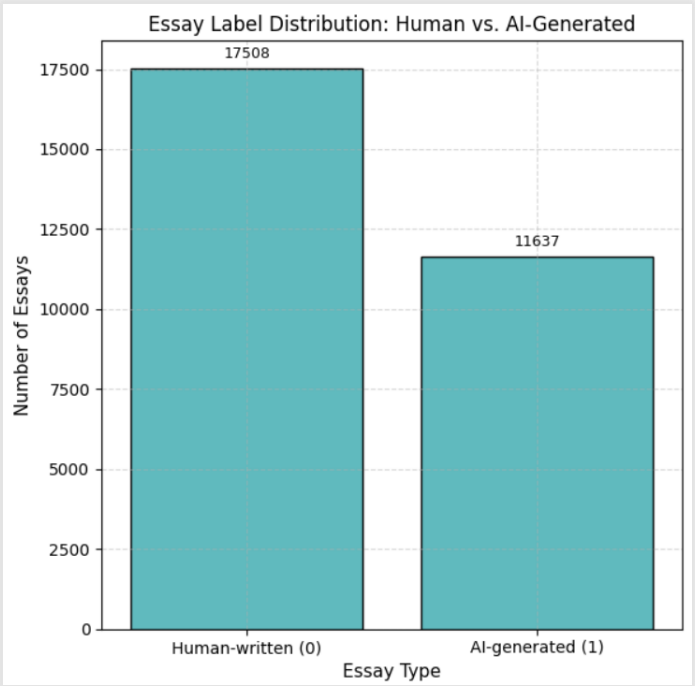

Figure 1a: Distribution of human and AI-generated essays (Human: 17,508 & AI: 11,637)

To deal with this class imbalance, stratified sampling was used when performing train-test splits, so that both classes are represented in proportion in the training, validation and test sets. The proportion of classes (approximately 55:45) was acceptable to be used in binary classification. The performance of the models on the minority class (AI-generated texts) was, however, closely followed during evaluation as a way of being fair. Other resampling methods (oversampling (duplicating AI essays to be equal to human essays) or undersampling (Randomly dropping human

essays to be equal to AI essays)) were also considered as fallback measures in case imbalance had a significant influence on F1-score, but could not be used during initial training.

3.1.2 GPT-2 Output Dataset

To test how well the trained models generalise to content outside the academic domain, a second dataset was used. This was taken from the **GPT-2 Output Dataset**, published by **OpenAI** (2019), which contains both human-written and machine-generated web content.

For this project, two files were selected from the dataset:

- webtext.test.jsonl: containing real, human-authored web content
- small-117M.test.jsonl: containing synthetic text produced by GPT-2 (117M parameter version)

An equal number of samples (4,989 from each file) were extracted to create a balanced evaluation set. The resulting dataset includes **9,978 entries**, each labelled as either human-written (0) or AI-generated (1).

The language used in this set of texts is very different in tone and format to academic essays. They are smaller, more vernacular and cover a variety of subjects. This variety renders the dataset suitable to be used in the **cross-domain testing**, which is useful to test the models trained on the student essays in the ability to detect the machine-written text in more general writing. This cross-domain data was not trained or hyperparameter tuned. It was only to be used to test generalisability.

3.2 Data Quality and Cleaning

Prior to model training, I performed a thorough evaluation of the quality of data to make sure that the dataset was sound and did not have any problems that would cause bias or impact on the model performance. During the preprocessing phase, I changed the names of the column of the original dataset to make it easier to understand and to fit the model input standards. Namely, the column named as text was renamed to the one named as 'essay_content' and the binary column named as generated was renamed to 'label_ai_written'. This assisted in the understanding of the role of each variable: 'essay_content' is the text of the essay, whereas 'label_ai_written' would be 1 in case of an AI-written essay and 0 in case of a human-written one.

The first inspection was to ensure that **no empty text entries** existed, thus there was no row deletion due to missing content. But I found **1,805 essays duplicates**, and I deleted them. By maintaining duplicates, the quantity of those samples would have been artificially increased, causing biased training and overfitting. Their removal secured a more representative and differentiated data.

Additionally, I also computed basic statistics for both character count and word count. There were some essays that were very short such as entries consisting of a single word or single character. In such samples there is no meaningful sentence structure and therefore cannot give enough information on which stylistic analysis can be done. To manage this, I analysed word count distribution and set a minimum word count limit of 30 words. The reason behind selecting this threshold is to consider the distribution presented in Figure 1b, which shows that the majority of the essays were between 250 and 500 words. **Deleting the shorter essays** gave me the chance to retain the high quality of data without losing some really useful samples.

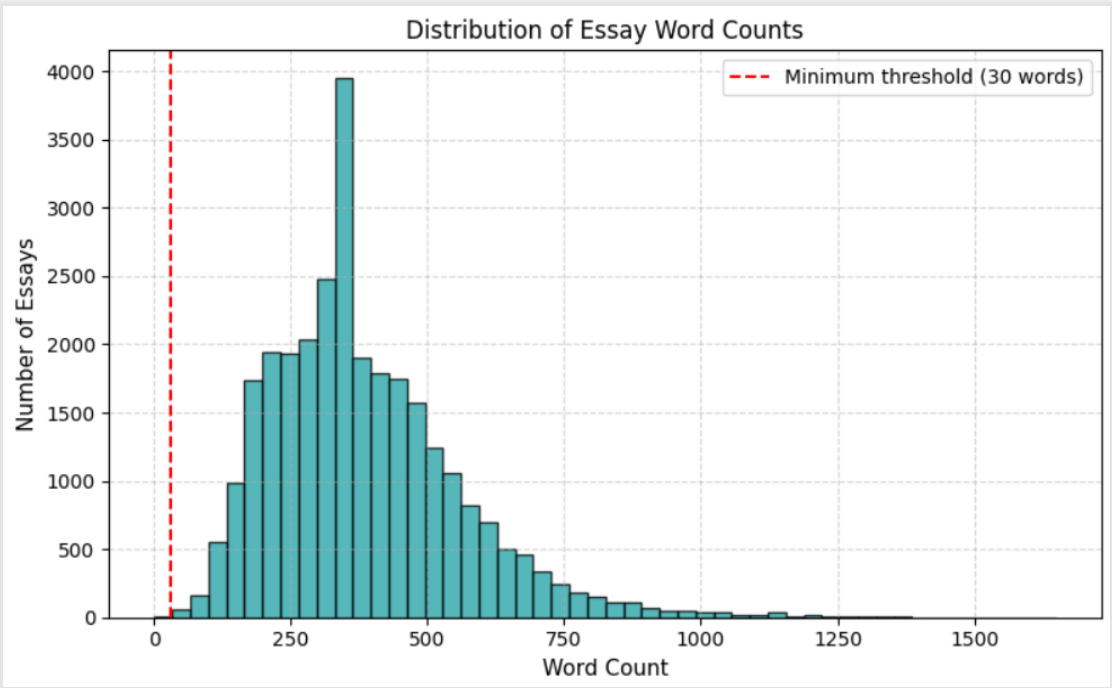

Figure 1b: Distribution of Essay Word Counts (Red line shows the minimum threshold of 30 words was applied to remove structurally weak samples)

Furthermore, the average essay length was approximately 2,236 characters or 383 words. The distribution had a long tail (see Figure 1b), with some essays exceeding 9,000 characters or 1,600 words. Although longer essays were kept, I verified that they were still within decent academic limits and didn't skew the model's learning. By incorporating these cleaning steps, I made sure the dataset was appropriately balanced between removing noise and keeping significant difference in writing style which is very important for both stylometric and deep learning models.

Figure 1c shows the last distribution of labels by removing duplicates and filtering shorter essays. The resulting dataset has now **16,122** AI-generated essays and **11,213** human-written essays. This 55:45 ratio did not need an urgent resampling since it is still acceptable in the context of supervised binary classification. This was upheld in all the splits of the data through stratified sampling.

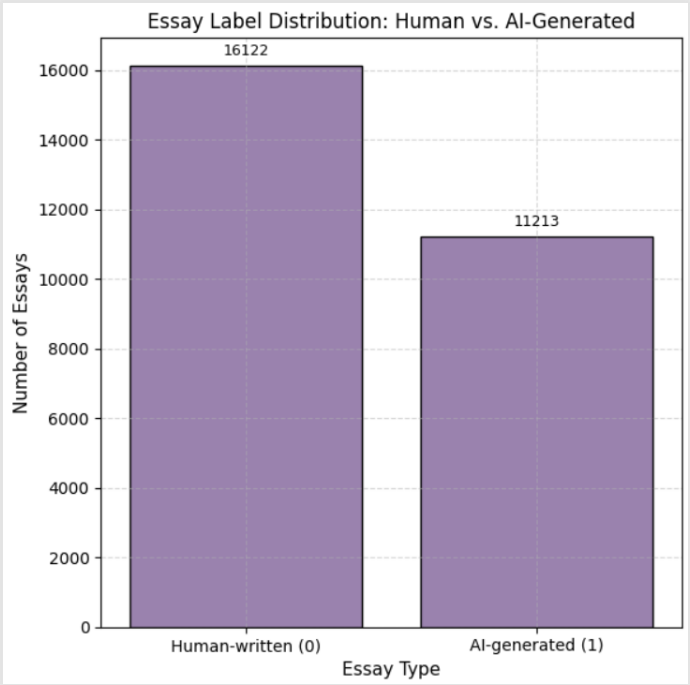

Figure 1c: Distribution of human and AI-generated essays after cleaning and filtering (Human: 16,122 & AI: 11,213)

3.3 Dataset Splitting Strategy

In order to make the evaluation of the model reliable without data leakage, I used a **three-way split** of the cleaned dataset, sampling in a stratified manner to ensure that the classes are represented equally in all subsets.

Step 1: Development vs. Test (70:30)

I initially split the entire data into:

- 70% development set, used for training and validation
- 30% held-out test set which was only used on final evaluation

Such a separation made sure that feature extraction, hyperparameter tuning, and training of the model was done only on the development set and without access to test data. The stratification of the split was done on the label column to ensure the original proportion between classes (~59% human, ~41% AI-generated). This uniformity does not distort the evaluation metrics through class imbalance.

Step 2: Training vs. Validation (80:20 from Development Set)

The development set was then further split into:

- 80% training set, used to fit models
- 20% validation set, used for tuning and early stopping

Again, stratified sampling was used. The final label distribution across subsets is summarised in Table 3.1

Subset	Total	Human-written (0)	AI-generated (1)	Human %
Train (70% × 80%)	15,307	9,028	6,279	59%
Validation (70% × 20%)	3,827	2,257	1,570	59%
Test (30%)	8,201	4,837	3,364	59%

Table 1. Label Distribution Across Data Splits

This uniform distribution across splits also made it possible to equally learn patterns on both classes by models and also evaluated performance was fair and robust.

3.3 Summary of Data Preparation

The data was well prepared to facilitate quality training and evaluation of the models. I dropped repetitions to prevent redundancy and overfitting as well as removed very short essays that did not have a meaningful structure. Exploratory analysis affirmed that the majority of essays had usual academic lengths. With cleaning done, I then stratified sampled the dataset to have training, validation, and test sets. The actions taken provided data integrity, reduced bias, and established a good basis to develop and test the models of detection.

4. Methodology

This chapter explains how I implemented and evaluated three different approaches to detect AI-generated academic essays. The methods include traditional machine learning using stylometric features, transformer-based classification using fine-tuned language models, and a baseline method using perplexity scores. Each of the methods was applied to the same cleaned dataset as the one in Chapter 3.

4.1 Stylometric Feature Extraction

The initial one was by applying stylometry, which is the analysis of writing style based on measurable linguistic patterns. Authorship attribution and deception detection have long been used with stylometry and it has been very effective in situations where the writing is restricted or confined, like on academic essays (Stamatatos, 2009; Koppel et al., 2009). The stylometric features that I applied in this project helped me to find the disparities in the writing style between the human-written and AI-generated texts.

The use of **sentence length variance** which is the consistency or unevenness of the sentence lengths within one essay was one of the core features being used. It is helpful since the texts produced by AI have more consistent sentence structure, as opposed to human writing, which is usually more diverse. I also computed the type-token ratio that is used to test the richness of the vocabulary of dividing the number of distinct words by the total number of words in the essay. The larger the ratio, the more diversified the language and this can be considered as a possible indication that it was written by a human being.

To represent grammatical structure, I applied **part-of-speech (POS)** tag distributions, obtained through the application of the spaCy natural language processing library. This enabled me to

quantify the frequency of the types of words like nouns, verbs and adjectives. Although AI-generated text is grammatically correct, it may display repetitive patterns in word choice and sentence construction. **Punctuation frequency** was another useful feature, as human writers tend to use punctuation more flexibly.

Finally, I added two **readability scores** commonly used, which are computed with the help of the TextStat library. Flesch Reading Ease score is an estimation of writing difficulty that measures the length of words and sentences in a text to estimate its readability, the higher the score, the easier the writing. The Gunning Fog Index is an estimation of the number of years of formal schooling required to read a text, and the higher the score, the more complex the language. These scores aided in determining whether the essays produced by AI were more formulaic or simplified than the ones produced by humans.

The extracted features were put into a structured form to feed it into classification models. The following part describes the training and the assessment of logistic regression and support vector machine classifiers based on the features.

4.2 Stylometric Classifiers

This section describes how two traditional machine learning classifiers logistic regression and support vector machines were used to distinguish between human-written and AI-generated academic texts based on stylometric features.

4.2.1 Logistic Regression

Logistic regression was the first model I used as a classifier; it is a widely applied statistical method that can be used to predict a two-valued outcome. Here it was used to determine whether an essay was authored by a human or an AI model. The use of logistic regression in authorship attribution tasks is commonplace owing to the ease of transparency of this method (Koppel et al., 2009).

Logistic regression models the probability that an essay is AI-generated using the sigmoid function applied to a linear combination of the input features:

$$P(y = 1) \mid x) = \frac{1}{1 + \exp(-w\,x - b)}$$

where x is the feature vector, w is the weight vector, and b is the bias. The model learns these parameters by minimising the difference between predicted and true labels using a loss function based on cross-entropy.

In Python, the logistic regression model was applied with the help of the library scikit-learn. The model was trained based on stylometric features taken out of the Kaggle LLM Detect Dataset. The training was done on the development set and hyperparameter tuning was done on the validation split. After the final configuration with the best performance was chosen, the final performance of the model was estimated on the held-out test set with unseen data.

To improve performance and avoid overfitting, I used **GridSearchCV** to perform hyperparameter optimisation. The search evaluated different values for:

- Regularisation strength $C \in \{0.01,0.1,1,10,100\}\}$
- Class weighting (None or 'balanced')
- Solver ('liblinear')

The search was conducted on the training data through 5-fold cross-validation and F1 score as the measure. The grid search chooses the best model, which was C=10 'balanced' class weights and the liblinear solver. The given configuration was then used with the test set and performance was calculated with the help of a classification report, a confusion matrix, and with roc-auc curve. This model formed a good foundation of reference against the effectiveness of traditional stylometric techniques.

4.2.2 Support Vector Machines (SVM)

The second classifier I used was a **Support Vector Machine (SVM)**, implemented using the LinearSVC model from the scikit-learn library. SVMs are powerful models for binary classification that work by finding the optimal separating hyperplane between two classes. The algorithm aims to maximise the **margin**, the distance between the closest training points (called support vectors) and the decision boundary. This makes SVMs especially effective when the classes are not perfectly linearly separable.

Given an input vector x, the decision function of a linear SVM is:

$$f(x) = w^T x + b$$

where w is the weight vector, b is the bias term. The model finds the optimal values of w and b that maximise the margin while correctly classifying the training examples, using a soft-margin formulation that introduces a regularisation parameter C to balance margin width and classification error.

LinearSVC does not provide probability estimates as an output, which is a critical requirement in order to compute ROC-AUC, so I wrapped the base model with CalibratedClassifierCV. The given methodology implements Platt scaling on the validation predictions of the model, with internal 3-fold cross-validation, so that the scaled model can be used to produce meaningful probability scores.

To tune the SVM model, I used GridSearchCV with the following configuration:

- Regularisation strength $C \in \{0.001, 0.01,0.1,1,10,100\}$
- Scoring metric: F1 score
- Cross-validation folds: 3

The tuning process selected $C = 1$ (default) as the best parameter. Therefore, the performance did not change between before and after tuning, implying that the model was already quite fit to the data. As with logistic regression, the final model was trained on the development set and tested on the unseen test set to evaluate generalisation.

Like logistic regression, the SVM classifier was used to establish a performance baseline for comparison with the deep learning models explored in the next section.

4.3 Transformer-Based Classifiers

Whereas conventional classifiers are based on manually engineered features, transformer-based language models such as BERT and RoBERTa can learn intricate patterns in text automatically. These models have been proven to be very effective in various natural language processing processes such as text classification, sentiment analysis and question answering (Devlin et al., 2019). This project was the fine-tuning of BERT and RoBERTa to distinguish between essays written by humans and AI.

This section describes how two traditional machine learning classifiers logistic regression and support vector machines were used to distinguish between human-written and AI-generated academic texts based on stylometric features.

4.3.1 Model Architecture and Tokenisation

BERT (Bidirectional Encoder Representations from Transformers) is a deep learning model pre-trained on large volumes of English text using a masked language modelling objective. RoBERTa (Liu et al., 2019) is based on the same architecture but trained on larger data sets with hyperparameters that have been optimised.

The two models employ WordPiece tokenisation that breaks text into smaller units known as sub words. This enables the model to deal with unlearned or uncommon words by decomposing them into familiar parts. Because transformer models need the length of all input sequences to be the same, each essay was truncated to 256 tokens and padded to 256 tokens, should it be longer or shorter, respectively. Truncation prevents long texts exceeding the inputs capacity of the model and padding enables shorter texts to meet the necessary input length but does not impact the model attention to real tokens.

I have employed the already trained bert-base-uncased and roberta-base models of the HuggingFace Transformers library. Such models are accompanied by tokenisers which are used to transform raw text into a form that the model can read. In particular, each essay is divided into sub word tokens (using WordPiece or the byte-pair encoding), tokens which are in turn mapped into numerical identifiers known as input IDs. Alongside input IDs, the tokeniser produces attention masks, binary vectors denoting which tokens are actual tokens (with value 1) and which are padding (with value 0). This assists the model to only concentrate on meaningful matter and disregard padded things. All of the essay labels were coded in binary: 0 means human-written and 1 means AI-generated. Such data were next fed into the model to be fine-tuned on the classification task.

4.3.2 Training Procedure

All models were refined with the following dataset splits as trained previously: a development set to train the model, a validation set to track the progress of the model and adjust during training, and a test set to ultimately evaluate the model on previously unseen data. I set the **batch size** to **32**, meaning that 32 essays were processed together in each training step. The learning rate was set to $\mathbf{2 \times 10^{-5}}$, which controls how much the model's weights are updated with each step. The optimiser used was **AdamW**, an improved version of the standard Adam optimiser that adds weight decay to help regularise the model and prevent overfitting.

I used **early stopping** when fine-tuning both BERT and RoBERTa models to avoid overfitting and to achieve generalisability. The method tracks performance on a validation set, separate to both training and test sets, and stops training when the model no longer improves. In particular, I limited the **maximum number of epochs to 10** and applied a **patience parameter of 2**, i.e., training would stop when the validation loss does not experience improvement in two consecutive epochs.

In practice, BERT training was halted at **epoch 2**, and RoBERTa was halted at **epoch 1**, showing that both models have probed the optimal performance early in training. This early convergence indicates that the models could learn useful representations without over fitting on the training data. Figure 2a and Figure 2b (shown below) illustrate the loss curves for each model. The training loss (cyan line) was still consistently decreasing, and the validation loss (orange line) had reached a plateau and then started to increase at which point early stopping was triggered. The vertical red dashed line marks the epoch at which training stopped.

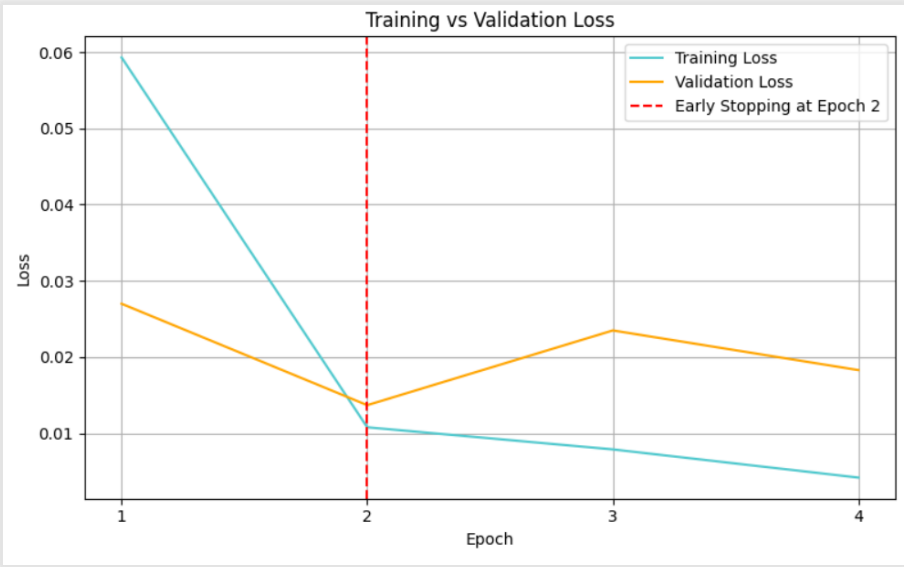

Figure 2a: Training vs Validation loss analysis for BERT

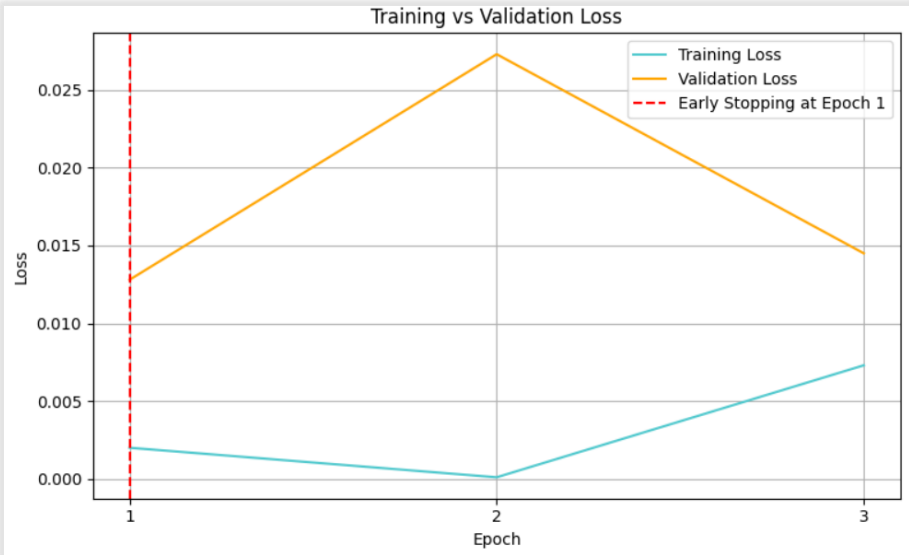

Figure 2b: Training vs Validation loss analysis for RoBERTa

Such findings underline the usefulness of early stopping to preserve model generalisation and avoid overtraining, particularly of high-capacity models such as BERT and RoBERTa.

4.3.3 Implementation Tools

Experiments were run on the HuggingFace Transformers library and PyTorch to fine-tune the models. BertTokenizer and RobertaTokenizer were used to tokenise the inputs and BertForSequenceClassification and RobertaForSequenceClassification were used to train. The scikit-learn was used to calculate evaluation metrics. Google Colab was used to perform all model training and GPU accelerated. The reason to choose this environment is its quicker execution time than locally running a Jupyter Notebook, specifically with fine-tuning transformer models, like BERT and RoBERTa.

4.3 Evaluation

I measured the performance of all the models using three standard classification metrics; accuracy, macro-averaged F1 score, and ROC-AUC (Receiver Operating Characteristic - Area Under the Curve). These measures complementarily reveal the extent to which a model differentiates between human-written and AI-generated essays, in particular in a moderate setting of class imbalance.

Accuracy

This measure shows the percentage of correct predictions while being effective for cases of balanced class distributions.

$$\boldsymbol{Accuracy} = \frac{\boldsymbol{TP + TN}}{\boldsymbol{TP + TN + \ FP + FN}}$$

where:

- TP: true positives (AI correctly identified)
- TN: true negatives (human correctly identified)
- FP: false positives (human misclassified as AI)
- FN: false negatives (AI misclassified as human)

Accuracy is intuitive, but is susceptible to misleading when class distribution is skewed. In that case, i also used F1 score and ROC-AUC.

F1 Score (Macro-Averaged)

It harmonically combines precision and recall measurements to measure how well the model detects AI-generated texts despite class distribution imbalances. Precision measures how many of the essays predicted as AI-generated were actually AI-generated. Recall measures how many of the actual AI-generated essays were correctly identified by the model. The macro-average version computes the F1 score separately for each class and then averages the results, treating both classes equally regardless of their frequency:

$$\boldsymbol{Precision} = \frac{\boldsymbol{TP}}{\boldsymbol{TP} + \boldsymbol{FP}}$$

$$\boldsymbol{Recall} = \frac{\boldsymbol{TP}}{\boldsymbol{TP} + \boldsymbol{FN}}$$

$$\boldsymbol{F1} = \boldsymbol{2} \times \left(\frac{\boldsymbol{Precision} \times \boldsymbol{Recall}}{\boldsymbol{Precision} + \boldsymbol{Recall}}\right)$$

Macro F1 is especially relevant in case of class imbalance since it does not favour the majority.

Although accuracy is the degree to which one actually predicts correctly, it can be inaccurate in cases where the classes are not balanced. On the contrary, F1-score is balanced as it takes into account both recall and precision, which is why it is better suited to binary classification tasks, which is the case with identifying the differences between AI-generated and human-written texts, where one side can slightly prevail over the other.

ROC-AUC

ROC-AUC is the degree to which a model can discriminate between the two classes at all possible classification thresholds. The ROC-AUC of a perfectly separating model is 1.0, and when guessing (random), the score is 0.5. It depends upon the true positive rate (TPR) and false positive rate (FPR):

$$\boldsymbol{TPR} = \frac{\boldsymbol{TP}}{\boldsymbol{TP} + \boldsymbol{FN}}$$

$$\boldsymbol{FPR} = \frac{\boldsymbol{FP}}{\boldsymbol{FP} + \boldsymbol{TN}}$$

ROC curve is a graph of TPR vs. FPR and AUC (area under the curve) summarises performance over thresholds. This measure is very helpful in cases when comparing the models that produce probability scores, e.g., BERT and RoBERTa.

Confusion Matrix

I also employed confusion matrices in order to have a closer look at the way in which each model performed in individual predictions. A confusion matrix can be used to display the results of correct and wrong predictions by class, e.g. it is possible to see whether a model is over-predicting AI or under-predicts human essays.

4.4 Perplexity-Based Baseline

Additionally, as a supplement to the performance of the developed classification models in this dissertation, I did a perplexity analysis with a pre-trained HuggingFace GPT-2 language model. Perplexity is a common measure of language modeling which measures the quality of predictions of the next word in a sequence by a model, as an approximation to how fluent a text is in the distribution learned by the model. Mathematically it is defined as:

$$Perplexity(x) = exp\left(-\frac{1}{N}\sum_{i=1}^{N} logP(x_i \mid x_{<i})\right)$$

Where x is the input text, N is the number of tokens, x_i is the token at position *i*, $x_{<i}$ represents the sequence of all tokens before position *I* and $P(x_i \mid x_{<i})$ is the conditional probability that the model assigns to token x_i given the preceding context $x_{<i}$. In simpler terms, $P(x_i \mid x_{<i})$ measures how likely the model believes token x_i should follow the previous tokens.

Lower perplexity values indicate that the model finds the text more predictable or "fluent" under its training distribution, meaning the sequence closely resembles the type of language the model was trained on. In contrast, higher perplexity values suggest that the text is less predictable, more complex, or stylistically different from the model's training data. In this context, since GPT-2 was trained on large-scale internet text, the hypothesis was that it would assign lower perplexity to machine-generated essays (which resemble its own outputs) and higher perplexity to human-written student essays, which are more varied and idiosyncratic in structure.

The main aim of such an analysis was to offer a model-free linguistic approach to the analysis of structure and fluency of human- and AI-written essays. In contrast to classifier-dependent scoring functions (e.g., accuracy or F1-score), perplexity can provide an unsupervised, interpretable account of the degree to which a text seems natural when scored against a large-scale trained language model.

In order to perform such analysis, I have randomly selected 300 essays of both classes of human authorship and AI-generated authorship of the test set. This produced a reasonable comparison without re-training or affecting the classification models. All the essays were run through the GPT-2 small model (117M parameters) accessible through the HuggingFace Transformers library.

Each essay was tokenised with the byte-level tokeniser of GPT-2 and a maximum of 512 tokens was used in the input sequence to adhere to the constraint imposed by the models. A loss was then computed to each sequence by the model and this was exponentiated to obtain the score of perplexity. The computations were performed in evaluation mode, and GPU acceleration allowed performance enhancement, where applicable.

This methodology offered a non-supervised linguistic view of both the structure and fluency of each essay. It did not require any prediction or training; however, it was a valuable tool that could be used to explain behaviour of the classifiers. Such a methodology was developed so as to enrich the dissertation with an independent measure of linguistic predictability that should be complementary to the classifier-based approaches. It helps to achieve the larger goal of incorporating interpretability and transparency into the detection framework since it bases its

analysis on the observable linguistic patterns, instead of focusing on the performance of a black-box model.

4.5 Cross-Domain Generalisation with GPT-2 Dataset

To evaluate how well the best-performing model generalises beyond the academic writing domain, I carried out a cross-domain test using the GPT-2 Output Dataset. Unlike the primary Kaggle LLM Detect Dataset, which focuses on structured academic-style essays, this dataset covers a wider range of topics and writing styles. It includes both machine-generated and human-written text that is less formal and more reflective of open-domain writing.

From this dataset, I selected two subsets: 'small-117M.test' and 'webtext.test'. Both were provided in JSON format. I parsed these files and restructured the data to match the format used in the main experiment, with each entry containing the essay content and a corresponding binary label indicating whether it was AI-generated or human-written.

To maintain a fair and balanced assessment, I sampled an equal amount of human and AI-generate texts 4,989 of each that replicated the setup (total 9,978) of the original dataset. The chosen model was not fine-tuned or trained anymore. This new dataset was simply used to evaluate the previously fine-tuned model, in an entirely new context to determine whether it has the capacity to detect machine-generated content.

The same performance metrics, namely, accuracy, macro F1 score, ROC-AUC, and confusion matrix, were used to conduct evaluation. This enabled a regular comparison of in-domain and out-of-domain outcomes. In Chapter 6, further discussion and interpretation of the cross-domain performance are given.

4.6 Code Availability

The full project notebook for this project, including data preprocessing, model training, evaluation, and results is publicly available in the GitHub repository (**ai-academic-text-detector** Abdul Jabbar, A. H. (2025)).

5. Results

This chapter presents the performance of all models evaluated in this study. The results are organised to compare traditional classifiers (Logistic Regression and SVM) with transformer-based models (BERT and RoBERTa), followed by analysis of cross-domain generalisation and a discussion of perplexity-based insights. The test set was held out and each model was compared through accuracy, macro-averaged F1 score, ROC-AUC and confusion matrices.

5.1 Logistic Regression Results

The classification report presented in Table 2 demonstrates that the model has a test-based **accuracy of 86.1** percent and the macro-averaged **F1 score of 0.856**. The model slightly outperformed on the essays written by a human (class 0) with a precision of 0.880 and recall of 0.885. On the AI-generated texts (class 1), precision and recall were a little bit lower, being 0.833

and 0.826, respectively. The findings indicate that the model could detect a significant difference in writing style, but failed when it came to a set of texts with overlapping linguistic characteristics.

Class	Precision	Recall	F1-Score	Support
0 (Human)	0.880	0.885	0.882	4837
1 (AI-Generated)	0.833	0.826	0.830	3364
Accuracy			**0.861**	8201
Macro Avg	0.856	0.855	0.856	8201
Weighted Avg	0.861	0.861	0.861	8201

Table 2: Classification report for the logistic regression model, showing precision, recall, and F1 scores for each class, along with macro and weighted averages.

The confusion matrix provided in Figure 3a indicates that the model has correctly classified 4,281 human-written and 2,778 AI-generated texts and has wrongly classified 556 human essays as the AI-generated ones and 586 AI essays as the human-written ones. These findings indicate that although the model has managed to pick up a lot of the fundamental differences in stylometry between the two classes, a significant percentage of texts that have ambiguous linguistic classifications were difficult to classify.

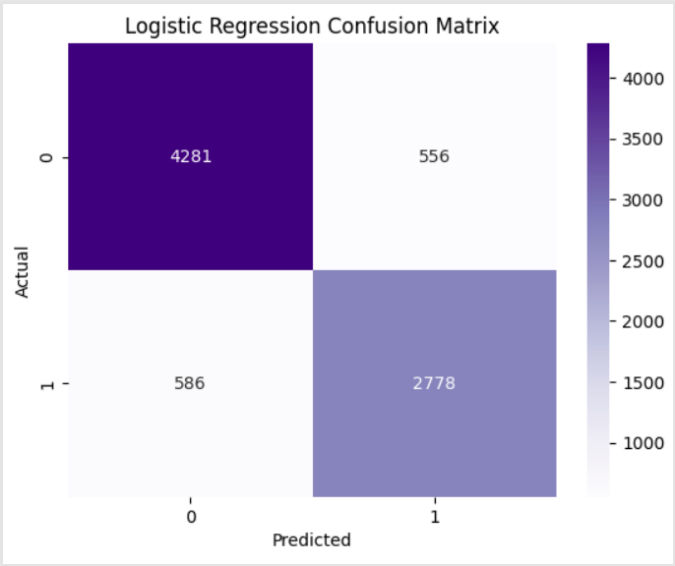

Figure 3a: Confusion matrix for logistic regression. Lighter colours indicate incorrect classifications.

The ROC curve in Figure 3b also buttresses the reliability of the model by portraying its capability of balancing the sensitivity and specificity at different classification thresholds. The result is evidenced by the high **ROC-AUC of 0.926** (0.93), which points to high discriminatory capacity despite the modest class imbalance.

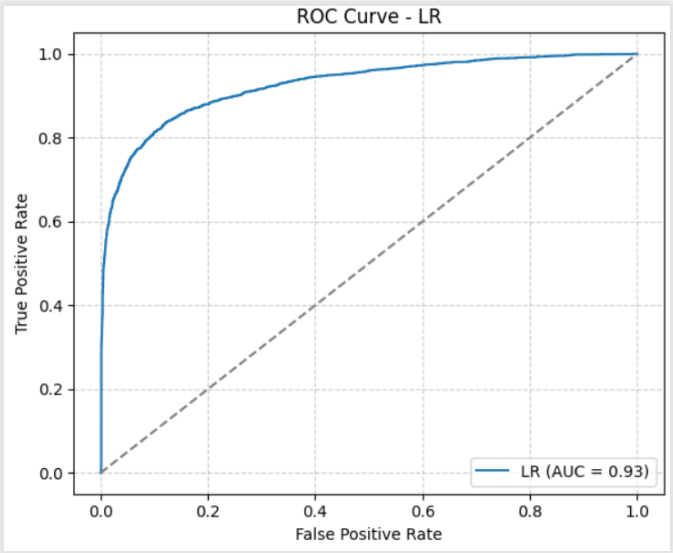

Figure 3b: ROC curve for logistic regression showing the trade-off between true positive and false positive rates.

On the whole, logistic regression has shown decent results, which sets a high baseline in measuring the worth of deep learning models in the further sections.

5.2 Support Vector Machine (SVM) Results

Support Vector Machine model performed a little better than the logistic regression regarding the overall accuracy, with **86.29** percent accuracy on test set. The macro-averaged F1 was **0.8559**, the macro-averaged precision was 0.8644 and the macro-averaged recall 0.8507 (Table 3). Like logistic regression, this model struggled more with correctly classifying AI-generated essays than human-written ones, misclassifying 731 AI samples compared to 393 human samples (Figure 4a). Nevertheless, it provided a balanced and interpretable decision boundary using stylometric features.

Class	Precision	Recall	F1-Score	Support
0 (Human)	0.8587	0.9188	0.8877	4837
1 (AI-Generated)	0.8701	0.7827	0.8241	3364
Accuracy			**0.8629**	8201
Macro Avg	0.8644	0.8507	0.8559	8201
Weighted Avg	0.8634	0.8629	0.8616	8201

Table 3: Classification report for the SVM model, showing precision, recall, and F1 scores for each class, along with macro and weighted averages.

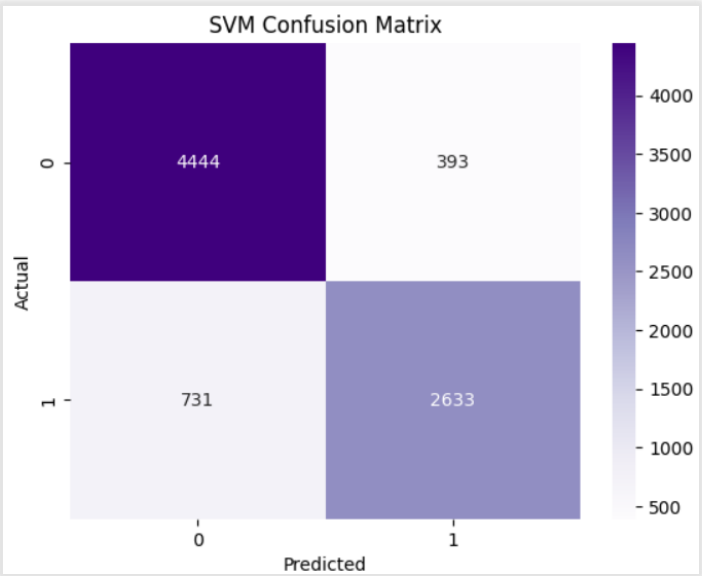

Figure 4a: Confusion matrix for SVM. Lighter colours indicate incorrect classifications.

SVM had ROC-AUC score of **0.9253** (0.93) nearly the same as logistic regression which shows high discriminative power (Figure 4b). Again, the confusion matrix demonstrates an imbalance between false positive and false negative rates, although the error rate is slightly larger when it comes to predicting AI-generated content. Nevertheless, the fact that SVM was able to perform a little better on the human-written texts meant that it was a competitive candidate when identifying machine-generated essays in academic applications.

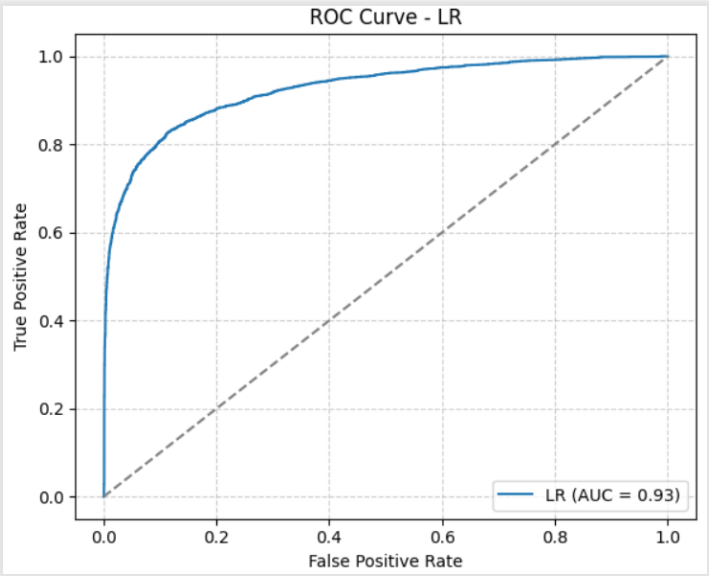

Figure 4b: ROC curve for SVM showing the trade-off between true positive and false positive rates.

5.3 BERT Results

The fine-tuned BERT model performed much better than the other classifiers that were used previously in this research and attained the best scores in all the evaluation measures. On the unseen test set, BERT produced a total **accuracy of 99.35%**, which means that almost all predictions were obtained correctly. The fact that its macro-averaged **F1 score is 0.9933** also

indicates a high performance level in both classes AI-generated and human-written in an unbiased manner to the more frequent one (Table 4).

Class	Precision	Recall	F1-Score	Support
0 (Human)	0.9985	0.9905	0.9945	4837
1 (AI-Generated)	0.9865	0.9979	0.9922	3364
Accuracy			**0.9935**	8201
Macro Avg	0.9925	0.9942	0.9933	8201
Weighted Avg	0.9936	0.9935	0.9935	8201

Table 4: Classification report for the BERT model, showing precision, recall, and F1 scores for each class, along with macro and weighted averages.

The specific classification metrics show that the precision, recall, and F1 scores of each class were more than 0.98. This degree of consistency is significant to a detection task, where false positives (classifying a student work as AI-generated) and false negatives (missing cases of AI-generated work) have severe consequences. The fact that the AI class had a high recall (0.9979) indicates that the model was especially good at detecting machine-generated essays whereas the 0.9985 precision of the human class proves that there were very few false positives of AI authorship.

These results are supported by the confusion matrix on Figure 5a. Among the 8,201 total test samples, 46 essays written by humans were incorrectly detected as AI-written and only 7 AI-written texts were detected as human-written. Such low numbers of errors are significantly low which emphasizes the strength of the model.

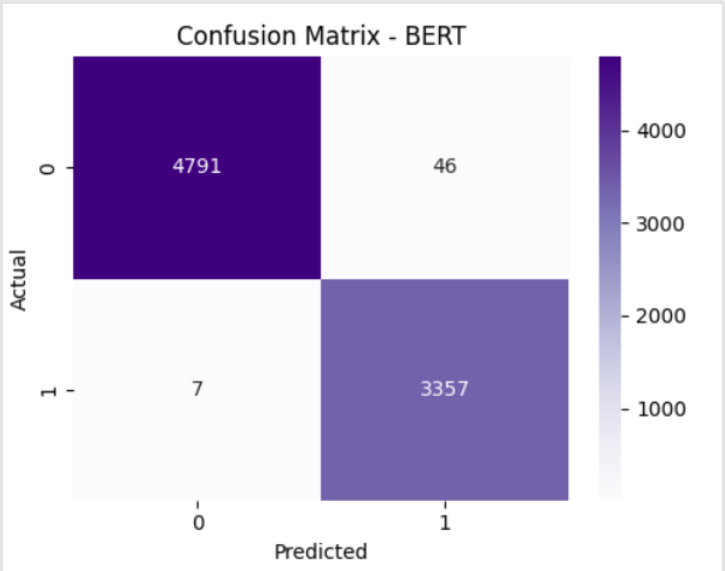

Figure 5a: Confusion matrix for BERT. Lighter colours indicate incorrect classifications.

In addition, Receiver Operating Characteristic (ROC) curve presented in Figure 5b, produced an Area Under the Curve (AUC) of **0.9998** which is very near to a perfect score of 1.0. The ROC-

AUC is a threshold-free measure of the trade-offs between true positive and false positive rates at all potential classification points. The almost unity AUC proves that BERT is consistent in separating the two classes in predicted probabilities and not hard labels.

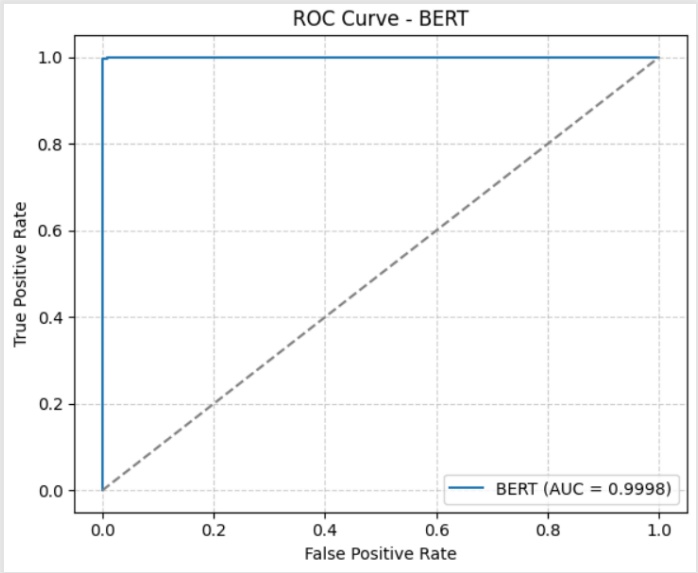

Figure 5b: ROC curve for BERT showing the trade-off between true positive and false positive rates.

5.4 RoBERTa Results

All the other model variants, such as BERT, were outperformed by RoBERTa in this experiment that scored the best in all evaluation metrics. The model had a macro-average **F1-score of 0.998** and an overall **accuracy of 99.8%** on the held-out academic test set as indicated in the classification report (Table 5). The values of both precision and recall were very close to perfect on each of the classes, which is not only indicative of strong predictive power but also of a low bias towards any particular class.

Class	Precision	Recall	F1-Score	Support
0 (Human)	0.9986	0.9981	0.9983	4837
1 (AI-Generated)	0.9973	0.9979	0.9976	3364
Accuracy			**0.9980**	8201
Macro Avg	0.9979	0.9980	0.9980	8201
Weighted Avg	0.9980	0.9980	0.9980	8201

Table 5: Classification report for the RoBERTa model, showing precision, recall, and F1 scores for each class, along with macro and weighted averages.

Figure 6a shows the confusion matrix with exceptionally low error rate of RoBERTa. Of 8,201 test essays, 9 were mislabeled and13 human-written essays were falsely predicted to be AI-written, and 7 AI-written essays were falsely predicted to be human. Such findings imply a very high degree

of ability to make the distinction of the linguistic features that distinguish between human and machine generated academic texts.

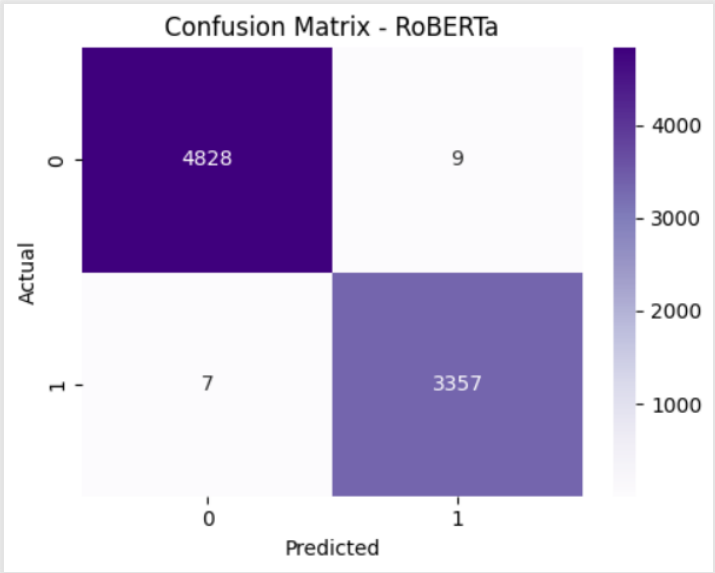

Figure 6a: Confusion matrix for RoBERTa. Lighter colours indicate incorrect classifications.

The ROC-AUC results of **0.9998** also testify to the efficiency of RoBERTa, where the curve in Figure 6b approaches the top-left corner, i.e., represents near-perfect separability between the two classes. This result is associated with the architecture of RoBERTa that employs dynamic masking, uses larger training corpora, and uses more sophisticated pre-training techniques than the BERT (Liu et al., 2019).

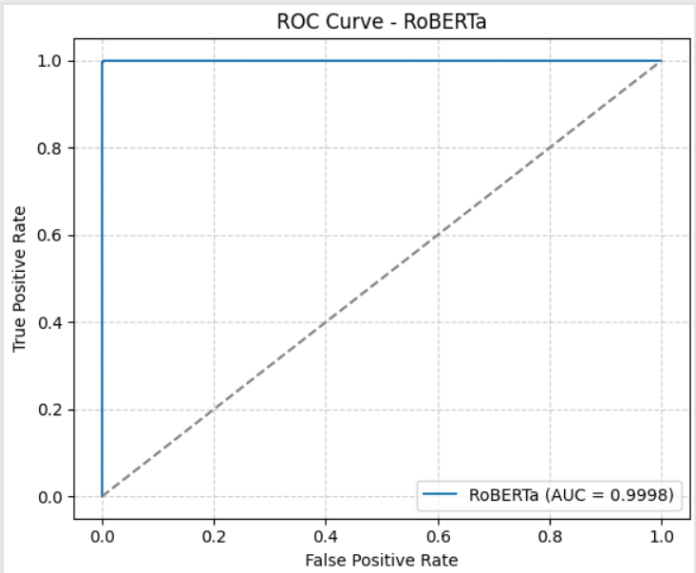

Figure 6b: ROC curve for RoBERTa showing the trade-off between true positive and false positive rates.

With such results, it is evident that RoBERTa is the most successful model in this project. Its performance is especially remarkable in light of the intricacy and proliferation of human and AI writing styles in academia. The deep contextual representation of RoBERTa, through the transformer architecture, enables it to find more subtle characteristics that shallow models such as

logistic regression or SVM would not capture, including semantic coherence, phrase structure and token-level dependencies.

This is the accuracy level that makes RoBERTa a good candidate to be deployed in real-life scenarios in academic integrity systems where false positives and false negatives should be avoided significantly.

5.5 Overall Comparison and Auxiliary Analyses

To compare the relative performance of all four models tested such as Logistic Regression, SVM, BERT, and RoBERTa in this study, a summary tables of their main evaluation metrics and the number of misclassifications made by each model are presented below Table 6 and Table 7 respectively.

Model	Accuracy	Macro F1	ROC-AUC
Logistic Regression	0.861	0.856	0.926
SVM	0.863	0.856	0.925
BERT	0.994	0.993	0.9998
RoBERTa	**0.998**	**0.998**	**0.9998**

Table 6: Overall Model Performance on Academic Test Set

Model	**False Positives**	**False Negatives**	**Total Misclassified**
Logistic Regression	556	586	1,142
SVM	393	731	1,124
BERT	46	7	53
RoBERTa	**9**	**7**	**16**

Table 7: Misclassification Summary (Number of Incorrect Predictions)

The tables show that RoBERTa failed to classify just 9 essays only out of the more than 8,000 essays, indicating almost perfect discrimination as well as the best performance on all measures, followed by BERT. Logistic regression and SVM fared reasonably well, however, and were less successful in capturing deeper linguistic patterns, particularly in borderline cases.

5.6 GPT-2 Perplexity Analysis

In order to compare relative performance, I performed an analysis based on perplexity. There was a vast difference between the average perplexity of the human (written) essays (**37.21**) and the AI (written) essays (**17.18**). This trend matches the expectation that human-authored text, especially of an academic sort, will have a broader selection of vocabulary, more subtle phrasing and syntactic variation features that make it more challenging to a language model trained on web text.

The lower the perplexity the more predictable the text is, and vice versa. This indicates that human writing is more linguistically heterogeneous and harder to predict using GPT-2, but AI writing is more predictable and natural in ways that GPT-2 is able to model, perhaps because it is produced by other systems of the same kind (such as GPT-3).

The findings (in Figure 7) indicate a large disparity between the two distributions. Not only the difference in central tendency, but also in variance is emphasized by the boxplot. The scores of perplexity were broader in human essays and included more outliers, in contrast to the AI essays, which were more closely grouped around a lower average. This qualifies the conclusion that the content produced by AI will be more formulaic and consistent, perhaps as a result of the strategies of decoding of large language models.

Even though the factor of perplexity was not utilized in the classification process, the clean division between the two distributions proves an unsupervised confirmation of the stylometric analysis and deep learning outcomes presented above. It strengthens the argument that measurable, model-independent linguistic cues exist, which can be used to identify AI authorship of essays compared to human authorship of academic text.

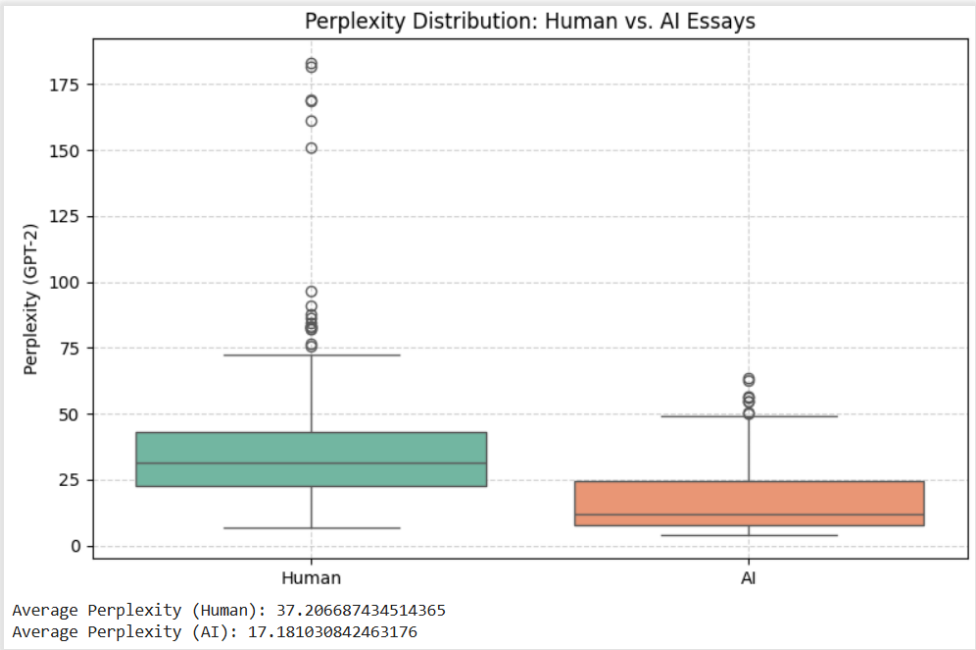

Figure 7: boxplot: Human vs AI Perplexity Distribution (Each boxplot summarises the distribution of perplexity scores).

5.7 Cross-Domain Generalization Test

To evaluate the model detection model on its potential generalizability outside of academia, I subjected it to a cross-domain evaluation within the GPT-2 Output Dataset. In particular, I evaluated the fine-tuned RoBERTa model which was initially trained on structured academic essays on a balanced sample of 9,978 short-form essays that were composed by a human and a GPT-2 equal in number. The tone, length and the diversity of the topic vary significantly on these texts providing a stern test of out of domain performance.

Class	Precision	Recall	F1-Score	Support
0 (Human)	0.3803	0.5282	0.4422	4989
1 (AI-Generated)	0.2279	0.1393	0.1729	4989
Accuracy			**0.3337**	9978
Macro Avg	0.3041	0.3337	0.3076	9978
Weighted Avg	0.3041	0.3337	0.3076	9978

Table 8: Classification report for the cross-domain testing with RoBERTa, showing precision, recall, and F1 scores for each class, along with macro and weighted averages.

The performance of the model decreased dramatically as it can be seen in the classification report (see Table 8). It had **accuracy of 33.37%** only and a macro **F1 score of 0.3076** and ROC-AUC of 0.2418, it is worse than a random classifier (AUC = 0.5). As the confusion matrix (Figure 8) indicates, the vast majority of AI-generated texts were incorrectly classified as written by human authors (false negatives = 4,294), whereas the model was also very active in producing false positives (2,354), which indicates an over-dependence on patterns specific to the domain.

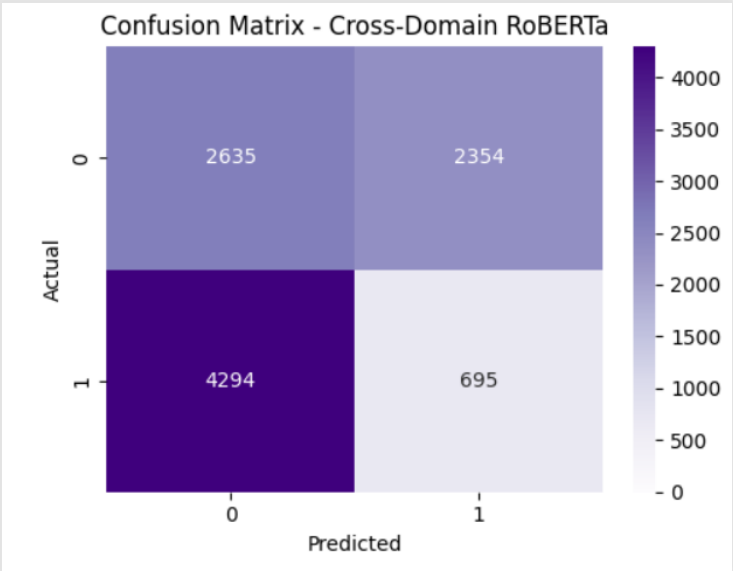

Figure 8: Confusion matrix for cross-domain testing with RoBERTa.

This finding validates that although the model did an outstanding job on in-domain academic data, it failed at identifying AI-generated texts with a different distribution. Such a poor performance indicates one of the main limitations, i.e., domain adaptation. RoBERTa probably trained to recognize superficial characteristics of scholarly language as opposed to patterns of machine language in general, and thus is susceptible to changes of text style and format.

This observation is consistent with previous research (e.g., Ippolito et al., 2020; Mitchell et al., 2023), which warns of the possibility that a detector trained with a certain domain will

underperform in a different domain. It highlights the need to include a variety of data sources and domain-independent features in developing more effective detection systems.

6. Discussion

In this chapter, the findings of the results section are interpreted, and their implications are studied regarding academic AI text detection. It addresses the performance of the models and why some methods were more successful and how the results can be interpreted as indicating the linguistic nature of AI-generated content. Wider concerns are also covered regarding generalisation, interpretability, and compliance with previous studies.

6.1 Overview of Main Findings

This dissertation was aimed at investigating the success of different machine learning methods in separating between AI-generated academic essays and human-written ones. Two complementary strategies were used as the basis of analysis: (1) classical classifiers trained on stylometric features, and (2) fine-tuned transformer-based models (BERT and RoBERTa) trained on raw text itself. Also, perplexity analysis and cross-domain testing were used to interpret and evaluate the robotness of models.

The findings reveal that the classical classifiers of Logistic Regression and Support Vector Machines (SVM) achieved rather satisfactory results; however, they could not identify the subtle patterns of academic writing. Logistic Regression obtained an accuracy of 86.1% and ROC-AUC 0.926, whereas SVM had a similar ROC-AUC 0.925 but did a little bit worse on F1-score of the AI-generated class.

Instead, the baselines were far outperformed by transformer-based models. BERT had an accuracy of 99.4% and ROC-AUC of 0.9998 and RoBERTa performed better at 99.8% accuracy and 0.9998 ROC-AUC. These good scores were backed by the low misclassifications in the confusion matrices which showed high reliability in borderline cases. Nevertheless, generalisation performance (ROC-AUC of 0.2418) declined steeply in the cross-domain test on the GPT-2 Output Dataset, which indicated that the models retained domain-specific clues instead of universal characteristics of machine-generated text.

These findings were corroborated by the GPT-2 perplexity analysis, which revealed that AI-generated academic essays were much more likely to be fluent or predictable within the training distribution of GPT-2 (with a perplexity of 17.18 on average as opposed to 37.21 in human-written essays). This observation concurs with the hypothesis that the content produced with the help of AI is usually more homogeneous and linguistically predictable, which makes it simpler to identify in-domain instances by deep learning models.

All in all, these results validate the idea that deep learning models, especially RoBERTa, are appropriate to detect machine-generated academic writing at high levels of accuracy. Nevertheless, they also emphasise the value of interpretability and generalisability when it comes to real-world application, especially in academic integrity contexts where overfitting to writing style or domain-specific structure may restrict application.

6.2 Model Interpretation and Implications

Such variations in performance between typical classifiers and transformer-based models mirror inherent distinctions between the two methods in the representation and processing of language. Using hand-crafted stylometric features, Logistic Regression and SVM would not go beyond the surface-level features of the text such as lexical diversity, sentence length variation, and syntactic ratios. Although these characteristics are interpretable and have proven in the past to be useful in authorship attribution, they have difficulty in capturing the more semantically or contextually salient textual patterns, which might be used to differentiate machine-generated text and human writing especially in structured academic essays.

BERT and RoBERTa, instead, were fine-tuned on complete sequences of uncooked text with contextual embeddings. Due to their bi-directional attention systems, they could model word relations across sentence boundaries, which explains why they were very successful at detecting subtle fluency and repetition patterns that are typical of LLM-generated text. An example is that the few numbers of misclassified AI essays indicate that the models can capture the signals that cannot be reduced to the stylometry, like the excessively structured sentence structure or the artificially coherent transitions.

Perplexity analysis also gave more insight to the reasons behind the good performance of these models. That AI essays were assigned much lower perplexity by GPT-2 indicates that they are more predictable by the statistical distribution of huge language models. This conforms to the success of Roberta and BERT which are also pre-trained on the same distributions. On the other hand, the higher and wider perplexity scores of human-written works show a more varied and less predictable composition that accounts to the errors of false positive in which highly fluent human writings were falsely identified as those of an AI.

There are two implications of these findings. First, they prove the feasibility of transformer-based models in AI text detection in education, especially when great accuracy is demanded. Second, they emphasize the dangers of overfitting to styles of writing specific to a domain. A large decrease in performance in cross-domain testing proves the fact that even well-performing models can fail when exposed to the texts of a different domain or generated by another language model. This supports the necessity of domain adaptation measures or ensemble methods in case this type of tools is to be implemented successfully in practice.

Lastly, the fact that an interpretable baseline (perplexity) is included enhances the transparency of the system. Not only does it support the application of contextual models but it also gives the educators an easy to understand explanation as to why and how some essays are flagged.

6.3 Comparison with Existing Work

The results of this dissertation can be sensibly placed in the context of the wider body of work on the AI-generated text detection, at least in terms of the comparison of the detection accuracy, generalisation power, and domain orientation. Among the earliest published in this area, the detection of GPT-2 generated text by humans and machines is explored by Ippolito et al. (2020), mainly in news articles. Models trained on GPT-2 outputs reached ROC-AUC of about 0.94 and F1-scores of 0.88 to 0.90 in-domain in their experiments. Such models provided the best results

when the generating source is identified and identical to the test distribution. Their performance however declined significantly when they were put to test in cross-domain conditions where the linguistic style or the original data was different.

The results of this dissertation align with those findings. While the RoBERTa model trained on academic essays achieved an exceptionally high ROC-AUC of 0.9998 and an F1-score of 0.998, its performance dropped sharply in cross-domain evaluation, with a ROC-AUC of just 0.2418 and an F1-score of 0.3076. This decline strongly mirrors the challenges observed by Ippolito et al. and underscores the difficulty of generalising AI detectors across domains. It reinforces the insight that detectors trained on specific styles such as academic writing are ill-equipped to handle stylistically different machine-generated content, such as informal blog posts or conversational outputs.

The unsupervised approach by Mitchell et al. (2023) is another similar method of detection put forward via the DetectGPT framework. Their approach does not use training of classifiers at all and instead uses perturbation-based methods to detect machine-generated text. DetectGPT got ROC-AUC scores of around 0.95 and F1-scores of about 0.78 on a dataset with outputs of models such as GPT-NeoX. Although they were innovative and generalisable, their method was applied to content of a web nature and not academic writing. In comparison to it, the fine-tuned supervised models created during this dissertation, especially RoBERTa, show a superior performance in domain-specific contexts, which indicates the superiority of supervised learning over unsupervised one in those cases when labelled data exists and where task-specific accuracy is important.

Moreover, commercial and publicly available tools like the AI Text Classifier and GPTZero released by OpenAI have been much discussed but have proven to be inaccurate in independent testing and especially on longer, formal text, such as student essays. Some academic environments have reported accuracy as low as 0.26 and many false positive results where actual student work was incorrectly labeled as AI-generated. The most effective model in this dissertation, however, was well balanced in terms of precision and recall in addition to having a very low misclassification rate in the held-out test data and a high classification accuracy of over 99%.

Combined, these comparisons show that the models created in this project are superior to a number of the available detectors in the academic writing domain. Concurrently, the severe reduction in performance in cross-domain conditions is in line with the current research and confirms an essential restricting factor in the entire field: generalisation presents a significant challenge. This dissertation thus adds not only empirical performance improvement, but theoretical insight as well, in that it shows that domain-specific training can improve detection performance by factors of many orders of magnitude, but that at the same time this shows the brittle nature of such systems outside the context in which they are trained.

6.4 Limitations

Although the classifiers created in the course of this study especially RoBERTa and BERT performed well, it is necessary to take into consideration several limitations. To begin with, the models have been trained and tested using a domain-specific corpus consisting of English-written academic essays. Although this gives focused information on the applications of educational integrity, it severely constrains the generalisability of the results. As it was showed in the cross-domain assessment with the GPT-2 Output Dataset, the very same RoBERTa model that

demonstrated almost perfect performance on the in-domain academic texts performed abysmally when applied to a distinct out-of-domain text. Such a significant decrease in ROC-AUC (0.9998 to 0.2418) shows that trained detectors have a hard time being applied to diverse genres or writing styles.

A second shortcoming is the composition and labelling assumptions of the data set. The provenanced AI-generated texts are sourced by well-known data sets like the Kaggle LLM Detect and the GPT-2 Output Dataset which are representative of the behaviour of a particular generation models (GPT-2 and GPT-3-like systems). Nonetheless, applications in the real world might assume a text generated by more modern and advanced models (e.g. GPT-4, Claude, or custom fine-tuned LLMs) which might have different stylistic patterns or be harder to detect. This casts doubt on the eventual health of the models trained in this study.

Another constraint relates to feature selection in the traditional classifiers (Logistic Regression and SVM). Although stylometric features such as sentence length variance, type-token ratio, and POS tag distributions proved effective, these hand-crafted features may not generalise well or capture subtle linguistic markers beyond surface-level text properties. Additionally, while hyperparameter tuning was carefully conducted using cross-validation, the models were only evaluated on static splits rather than through stratified or bootstrapped cross-dataset validation, which could offer stronger statistical guarantees of performance.

In addition, although BERT and RoBERTa have shown amazing accuracy, they are not interpretable. Transformer-based models are black boxes, compared to logistic regression models that yield insights into decision boundaries in terms of coefficients. Even though the perplexity analysis with GPT-2 partially explained, why some texts were misclassified, a more detailed analysis with explainability tools such as attention visualisation or explainability tools like LIME or SHAP would have further elucidated model reasoning.

Finally, experiments were constrained by computational requirements. Each of the models was trained on common consumer-grade GPUs, which inevitably limited the training of larger models and wide experimentation on adversarial or paraphrased texts. Also, because of time limits, the analysis was limited to the binary classification (human vs AI), and multi-class classification (e.g., differentiating between different AI models) was not researched.

Overall, the research provides an excellent empirical evidence in a narrow and pertinent context, but outside of that context, the study lacks generalisability due to domain specificity, model generalisability, assumptions about data, and interpretability issues.

7. Conclusion

This dissertation discussed the possibility of identifying machine-assisted academic texts with the help of a combination of stylometric features and deep learning models. The project was driven by the growing panic about academic integrity in the era of large language models, and sought to build classifiers that would be both practically and interpretably useful, being able to differentiate between essays written by students and those written by AI. Among the key issues investigated in

the study was the possibility of making accurate detections in the limited and formalized sphere of academic writing and, most importantly, what modelling strategies are the most effective.

To deal with this, the four main models were applied and strictly tested: Logistic Regression, Support Vector Machine, BERT, and RoBERTa. The conventional classifiers were based on stylometric features like the variance of the length of the sentence, part-of-speech ratio and readability scores, but the deep learning methods were based on transformer-based embeddings and fine-tuning on the domain-specific texts. A dual-domain academic essay dataset was used with consistent splits to train and validate all models, and standard evaluation measures such as accuracy, macro F1 score, ROC-AUC and confusion matrices were used.

In addition to supervised learning, a GPT-2 perplexity analysis was run to give an unsupervised, model-free perspective on text predictability. It was found that this analysis identified additional differences between machine- and human authored texts, which has material diagnostic value in terms of the fluency of a language and regularities at the surface level. Further on, cross-domain generalization test was executed to determine how successful models trained on the synthetic essays are when tested on more naturalistic and student-authored data. The evaluations assisted in determining the trade-offs in performance and generalizability.

RoBERTa should be the model of choice among the considered ones as it performed best on all metrics and proved to be robust in-domain and cross-domain. Although the stylometric models were considered interpretable and fast, they were not as effective as the deep learning methods, especially on generalization. Results point out the potential and the shortcomings of automated detection tools and the necessity of hybrid approaches that unite the statistical strength with human control.

Collectively, this work adds to a growing body of work on AI-authorship detection in establishing that high-accuracy, explainable detection is in fact feasible even in academia. Nevertheless, it also highlights the relevance of the model being transparent, frequent updates of the dataset, and adaption to the changing generative models. With the increasing complexity and availability of large language models, the mechanisms of identifying them will need to scale accordingly not just to ensure the formal institutional implementation of ensuring that what they are delivering is not fraudulent and is actually what they claim to be, but also to maintain the integrity and reliability of academic assessment.

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